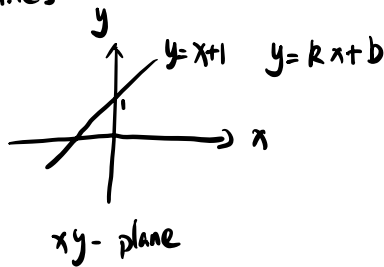
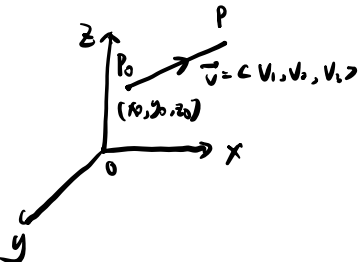


Lecture 3 Parametric Equations

Lines



time variable t



$$\langle \vec{r}, \vec{r} \rangle = \langle \vec{r}_0, \vec{r}_0 \rangle = t \langle v_1, v_2, v_3 \rangle$$

Vector form: $\vec{r}(t) = \vec{r}_0 + t\vec{v}$

Equation form:
$$\begin{cases} x = x_0 + tv_1 \\ y = y_0 + tv_2 \\ z = z_0 + tv_3 \end{cases}$$

Example. Find parametric equation of line L passing through both $A(3, -2, 0)$ and $B(1, 0, 1)$.

Sol. Let P be $A(3, -2, 0)$

$$\vec{AB} = \vec{OB} - \vec{OA} = \langle 1, 0, 1 \rangle - \langle 3, -2, 0 \rangle = \langle -2, 2, 1 \rangle$$

vector form of L : $\vec{r}(t) = \langle 3, -2, 0 \rangle + t \langle -2, 2, 1 \rangle$

equation form:
$$\begin{cases} x = 3 - 2t \\ y = -2 + 2t \\ z = 0 + t \end{cases}$$

Plane.

$z = 0$ represents xy -plane in the xyz -space

$$Ax + By + Cz = 1 \quad \begin{matrix} (x_1, y_1, z_1) \\ (x_2, y_2, z_2) \\ (x_3, y_3, z_3) \end{matrix} \Rightarrow \begin{cases} Ax_1 + By_1 + Cz_1 = 1 \\ Ax_2 + By_2 + Cz_2 = 1 \\ Ax_3 + By_3 + Cz_3 = 1 \end{cases} \Rightarrow \begin{cases} A = \\ B = \\ C = \end{cases}$$

3 points $\Rightarrow$ a plane

1 point + normal vector $\Rightarrow$ a plane
 $P_0(x_0, y_0, z_0)$ $\vec{n} = \langle A, B, C \rangle$

For any point $P(x, y, z)$ on this plane, we have

$$\vec{n} \cdot \vec{P_0P} = 0$$

$$\langle A, B, C \rangle \cdot (\langle x, y, z \rangle - \langle x_0, y_0, z_0 \rangle) = 0$$

$$Ax + By + Cz = Ax_0 + By_0 + Cz_0$$

$$Ax + By + Cz = D$$

$P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$

$\vec{n} = \langle A, B, C \rangle$ is the normal vector

$$\begin{aligned} \vec{P_1P_2} \cdot \vec{n} &= \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle \cdot \langle A, B, C \rangle \\ &= (Ax_2 + By_2 + Cz_2) - (Ax_1 + By_1 + Cz_1) \\ &= 0 \end{aligned}$$

Example. Find an equation of the plane in $\mathbb{R}^3$ passing through $A(0, 2, -1)$, $B(4, 0, -1)$ and $C(7, -3, 0)$.

Sol. $A(0, 2, -1)$ is on the plane.

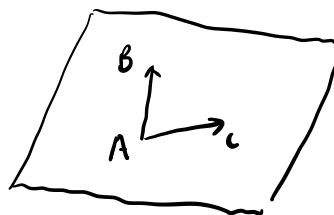
$$\vec{AB} = \langle 4, 0, -1 \rangle - \langle 0, 2, -1 \rangle = \langle 4, -2, 0 \rangle$$

$$\vec{AC} = \langle 7, -3, 0 \rangle - \langle 0, 2, -1 \rangle = \langle 7, -5, 1 \rangle$$

$$\vec{AB} \times \vec{AC} = \langle -2, -4, -6 \rangle$$

$$\vec{n} = \langle 1, 2, 3 \rangle$$

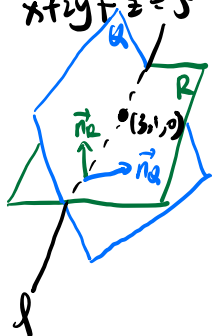
$$\text{Equation of the plane: } x + 2y + 3z = 1 \cdot 0 + 2 \cdot 2 + 3 \cdot (-1) = 1$$



Example. Find an equation of the line of intersection of the planes

$$Q: x + 2y + z = 5 \quad \text{and} \quad R: 2x + y - z = 7.$$

Sol.



$$\text{Let } z=0 \quad \begin{cases} x+2y=5 \\ 2x+y=7 \end{cases} \Rightarrow \begin{cases} x=3 \\ y=1 \end{cases}$$

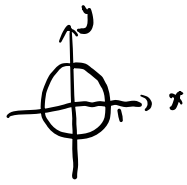
$\Rightarrow (3, 1, 0)$ is on the line l

$$\vec{n}_Q = \langle 1, 2, 1 \rangle, \quad \vec{n}_R = \langle 2, 1, -1 \rangle$$

$$\vec{n}_Q \times \vec{n}_R = \langle -3, 3, -3 \rangle$$

$$\begin{aligned} \text{Vector form of } l: \vec{r}(t) &= \langle 3, 1, 0 \rangle + t \langle -3, 3, -3 \rangle \\ &= \langle 3-3t, 1+3t, -3t \rangle. \end{aligned}$$

Curves



$$x^2 + y^2 = 1 \quad \text{Cartesian equation}$$

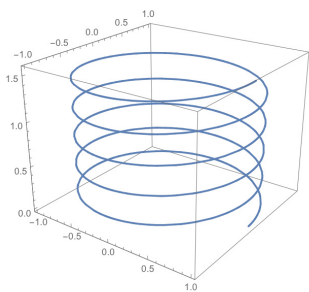
$$\begin{cases} x = \cos t \\ y = \sin t \\ z = 0 \end{cases}$$

Parametric equation

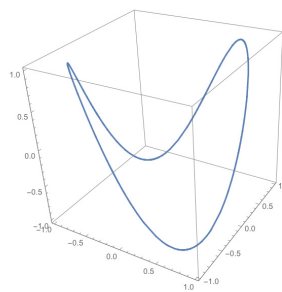
Parametric equation of a curve

$$\begin{cases} x = f(t) \\ y = g(t) \\ z = h(t) \end{cases}$$

$$\vec{r}(t) = f(t) \vec{i} + g(t) \vec{j} + h(t) \vec{k}$$



(a) sketch of $\vec{r}_1(t)$, a helix



(b) sketch of $\vec{r}_2(t)$

$$\vec{r}_1(t) = (\cos t) \vec{i} + (\sin t) \vec{j} + \frac{t}{20} \vec{k}$$

$$\vec{r}_2(t) = (\sin t) \vec{i} + (\cos t) \vec{j} + (\sin 2t) \vec{k}$$

Derivatives of Curves,

$\vec{r}(t) = f(t) \vec{i} + g(t) \vec{j} + h(t) \vec{k}$ represents the position of a particle at time t

$\vec{v}(t) \vec{r}'(t) = f'(t) \vec{i} + g'(t) \vec{j} + h'(t) \vec{k}$ represents the velocity vector of the particle

$|\vec{r}'(t)| = \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2}$ is the speed of the particle.

$\vec{a}(t) \vec{r}''(t) = f''(t) \vec{i} + g''(t) \vec{j} + h''(t) \vec{k}$ represents the acceleration vector of the particle

Newton's second law $\vec{F} = m \vec{a} = m \vec{v}' = m \vec{r}''$

$\downarrow$ $\downarrow$
 force mass

Example. $\vec{r}(t) = (\sin t) \vec{i} + (t^2 - \cos t) \vec{j} + e^t \vec{k}$

Find the velocity, speed and acceleration of the particle.

Sol. velocity = $\vec{r}'(t) = (\cos t) \vec{i} + (2t + \sin t) \vec{j} + e^t \vec{k}$

$$\begin{aligned} \text{speed} &= |\vec{r}'(t)| = \sqrt{\cos^2 t + 4t^2 + 4t \sin t + \sin^2 t + e^{2t}} \\ &= \sqrt{1 + 4t^2 + 4t \sin t + e^{2t}} \end{aligned}$$

acceleration = $\vec{r}''(t) = (-\sin t) \vec{i} + (2 + \cos t) \vec{j} + e^t \vec{k}$